

MORTLAKE, AustraliaWMO: **948400**Lat: **38.0736S**Lon: **142.7744E**Elev: **430**StdP: **14.47**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.7	35.6	31.3	26.0	46.0	33.6	28.7	43.3	29.0	51.6	26.7	52.4	3.6	0	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.0	94.2	65.5	89.2	64.6	84.3	63.8	69.2	83.0	67.3	81.2	65.7	79.0	13.3	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	95.5	71.2	63.3	88.5	70.0	61.5	82.9	68.5	33.6	83.0	32.0	81.0	30.7	78.7	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	22.6	20.3	DB	28.9	103.8	2.0	4.3	27.5	106.9	26.3	109.5	25.2	111.9	23.7	115.1	(4)
(5)			WB	28.7	73.0	1.9	1.8	27.3	74.3	26.2	75.4	25.1	76.4	23.7	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	64.4	65.4	61.9	57.1	52.3	48.5	47.6	48.4	50.6	53.5	57.4	60.9	(6)	
	DBStd	8.51	7.73	7.70	7.11	5.87	4.44	3.55	3.30	3.87	4.47	5.95	6.98	7.32	(7)	
	HDD50	336	0	0	0	5	23	70	89	78	42	24	5	0	(8)	
	HDD65	3758	106	82	151	250	394	496	539	515	432	363	252	178	(9)	
	CDD50	2383	448	431	370	216	95	24	15	28	61	131	227	337	(10)	
	CDD65	327	89	93	56	11	0	0	0	0	0	5	23	50	(11)	
	CDH74	4515	1249	1142	712	153	3	0	0	0	3	110	368	775	(12)	
	CDH80	2045	636	550	303	37	0	0	0	0	0	27	129	363	(13)	
(14) Wind	WSAvg	9.5	9.6	9.4	9.0	8.5	9.0	8.9	10.2	10.5	10.3	9.9	9.5	9.6	(14)	
(15) Precipitation	PrecAvg	27.0	1.4	1.2	1.6	2.2	2.6	2.5	3.0	3.1	2.8	2.6	2.3	1.9	(15)	
	PrecMax	37.6	5.6	3.4	3.8	6.1	5.9	5.2	6.0	5.1	5.2	5.9	4.5	5.3	(16)	
	PrecMin	16.0	0.1	0.0	0.3	0.2	0.4	0.6	1.0	0.6	0.8	0.5	0.4	0.1	(17)	
	PrecStd	4.5	1.1	1.0	0.9	1.5	1.3	1.0	1.1	1.0	1.1	1.2	1.0	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.4	98.9	95.2	84.9	72.2	62.4	60.5	66.0	73.0	83.6	89.9	98.3	(19)	
		MCWB	67.8	66.5	65.2	60.9	56.8	55.3	53.3	56.8	61.2	62.5	64.6	64.7	(20)	
	2%	DB	93.6	93.0	87.9	78.2	66.9	59.0	57.4	61.5	66.9	76.3	83.8	89.9	(21)	
		MCWB	65.7	65.9	63.5	59.3	56.2	53.8	51.9	53.6	57.4	60.9	63.8	63.6	(22)	
	5%	DB	87.3	87.2	82.2	73.4	63.2	57.0	55.5	58.2	63.0	70.6	77.9	82.7	(23)	
		MCWB	65.2	65.1	62.5	58.2	55.4	52.8	50.9	52.2	55.4	59.1	63.0	63.2	(24)	
	10%	DB	80.3	80.9	76.0	68.3	60.4	55.1	53.8	55.8	59.8	65.5	71.4	76.1	(25)	
		MCWB	63.4	64.1	61.5	57.3	54.8	51.7	49.9	50.8	53.6	57.0	60.7	61.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.5	72.1	68.0	63.8	60.3	57.3	55.4	57.7	62.0	65.6	69.6	69.3	(27)	
		MCDB	87.4	85.4	84.0	76.5	64.8	59.2	58.3	64.2	69.7	77.8	81.1	82.5	(28)	
	2%	WB	69.1	69.0	65.8	61.7	58.4	55.0	53.1	54.7	58.5	62.0	66.3	66.7	(29)	
		MCDB	83.7	82.6	79.1	71.2	63.5	57.6	55.9	59.5	64.8	72.4	78.4	80.3	(30)	
	5%	WB	66.9	67.0	64.2	60.0	56.8	53.5	51.7	52.9	56.2	59.8	63.8	64.6	(31)	
		MCDB	81.6	80.7	76.1	68.7	61.7	56.0	54.5	57.1	62.1	69.3	74.1	77.9	(32)	
	10%	WB	64.7	65.1	62.5	58.5	55.3	52.1	50.5	51.3	54.1	57.5	61.3	62.5	(33)	
		MCDB	79.0	77.7	73.7	66.3	59.5	54.5	53.1	54.9	58.9	64.3	70.9	75.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	25.4	24.0	22.7	20.0	15.2	13.4	12.9	14.5	16.6	19.9	21.6	23.9	(35)	
		MCDBR	38.0	34.0	32.9	28.6	20.2	15.9	14.6	18.1	21.9	29.9	32.5	36.2	(36)	
		MCWBR	12.7	10.9	11.4	12.3	11.7	11.3	10.1	11.5	13.8	16.0	14.7	13.6	(37)	
	5% WB	MCDWR	29.9	27.3	26.9	22.5	16.7	13.7	13.0	16.4	19.9	26.5	28.1	29.1	(38)	
		MCWBR	12.5	10.9	11.3	11.2	10.8	10.4	9.6	11.3	13.3	15.4	14.4	13.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.339	0.325	0.313	0.305	0.296	0.294	0.306	0.331	0.337	0.340	0.335	(40)	
	taud		2.467	2.504	2.543	2.557	2.560	2.569	2.569	2.524	2.435	2.425	2.451	2.490	(41)	
	Ebn at Noon		313	308	299	283	266	259	267	281	292	305	313	318	(42)	
	Edh at Noon		37	34	30	26	23	21	22	26	33	36	37	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2278	1969	1545	1073	714	568	625	891	1267	1693	1990	2240	(44)	
(45)	RadStd		135	101	80	71	34	43	39	50	81	130	125	137	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.68	N/A	N/A	N/A	N/A	+1.44	N/A	N/A	+205	N/A	(46)
(47)	Regional (31 neighbors)	+0.83	N/A	N/A	+1.57	+0.72	N/A	N/A	-230	+293	+86	(47)

Nomenclature: See separate page